

● HARD

● ADVANCED MATH

● ANSWER KEY

Advanced Math

16

10 answers · Rationale-rich · Teach from doc

Q1**B**

B. $n = 722.5^t$

Choice B is correct. Since the model estimates that the number of squirrels in the population increased by a fixed percentage, 150%, each year, the model can be represented by an exponential equation of the form $n = a1 + \frac{p}{100}^t$, where a is the estimated number of squirrels in the population at the end of 2015, and the model estimates that at the end of each year, the number is $p\%$ more than the number at the end of the previous year. Since the model estimates that at the end of each year, the number was 150% more than the number at the end of the previous year, $p = 150$. Substituting 150 for p in the equation $n = a1 + \frac{p}{100}^t$ yields $n = a1 + \frac{150^t}{100}$, which is equivalent to $n = a1 + 1.5^t$, or $n = a2.5^t$. It's given that the estimated number of squirrels at the end of 2016 was 180. This means that when $t = 1$, $n = 180$. Substituting 1 for t and 180 for n in the equation $n = a2.5^t$ yields $180 = a2.5^1$, or $180 = 2.5a$. Dividing each side of this equation by 2.5 yields $72 = a$. Substituting 72 for a in the equation $n = a2.5^t$ yields $n = 722.5^t$.

Choice A is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was 150% of, not 150% more than, the estimated number at the end of the previous year.

Choice C is incorrect. This equation represents a model where at the end of each year, the estimated number of squirrels was 150% of, not 150% more than, the estimated number at the end of the previous year, and the estimated number of squirrels at the end of 2015, not the end of 2016, was 180.

Choice D is incorrect. This equation represents a model where the estimated number of squirrels at the end of 2015, not the end of 2016, was 180.

Q2**C**

C. $fx = x^2 + 91x$

Choice C is correct. It's given that the function f gives the product of a number, x , and a number that is 91 more than x . A number that is 91 more than x can be represented by the expression $x + 91$. Therefore, f can be defined by the equation $f(x) = x(x + 91)$, or $f(x) = x^2 + 91x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3

B

B. a

Choice B is correct. A cube has 6 faces of equal area, so if the total surface area of a cube is $6\left(\frac{a}{4}\right)^2$, then the area of one face is $\left(\frac{a}{4}\right)^2$. Likewise, the area of one face of a cube is the

square of one of its edges; therefore, if the area of one face is $\left(\frac{a}{4}\right)^2$, then the length of one edge of the cube is $\frac{a}{4}$. Since the perimeter of one face of a cube is four times the length of one edge, the perimeter is $4\left(\frac{a}{4}\right) = a$.

Choice A is incorrect because if the perimeter of one face of the cube is $\frac{a}{4}$, then the total surface area of the cube is $6\left(\frac{\frac{a}{4}}{4}\right)^2 = 6\left(\frac{a}{16}\right)^2$, which is not $6\left(\frac{a}{4}\right)^2$. Choice C is incorrect

because if the perimeter of one face of the cube is $4a$, then the total surface area of the cube is $6\left(\frac{4a}{4}\right)^2 = 6a^2$, which is not $6\left(\frac{a}{4}\right)^2$. Choice D is incorrect because if the perimeter of one face

of the cube is $6a$, then the total surface area of the cube is $6\left(\frac{6a}{4}\right)^2 = 6\left(\frac{3a}{2}\right)^2$, which is not $6\left(\frac{a}{4}\right)^2$.

Q4

B

B. $-\frac{7}{4}, 0$

Choice B is correct. Since the line of symmetry for the graph of a quadratic function contains the vertex of the graph, the x -coordinate of the vertex of the graph of $y = f(x)$ is the x -coordinate of the midpoint of its two x -intercepts. The midpoint of two points with x -coordinates x_1 and x_2 has x -coordinate x_m , where $x_m = \frac{x_1 + x_2}{2}$. It's given that the vertex is $-3, 6$ and one of the x -intercepts is $-\frac{17}{4}, 0$. Substituting -3 for x_m and $-\frac{17}{4}$ for x_1 in the equation $x_m = \frac{x_1 + x_2}{2}$ yields $-3 = \frac{-\frac{17}{4} + x_2}{2}$. Multiplying each side of this equation by 2 yields $-6 = -\frac{17}{4} + x_2$. Adding $\frac{17}{4}$ to each side of this equation yields $-\frac{7}{4} = x_2$. Therefore, the other x -intercept is $-\frac{7}{4}, 0$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

A

A. 0, 36

Choice A is correct. It's given that the table shows values of x and their corresponding values of gx , where $gx = \frac{fx}{x+3}$. It's also given that f is a linear function. It follows that an equation that defines f can be written in the form $fx = mx + b$, where m represents the slope and b represents the y -coordinate of the y -intercept 0, b of the graph of $y = fx$ in the xy -plane. The slope of the graph of $y = fx$ can be found using two points, x_1, y_1 and x_2, y_2 , that are on the graph of $y = fx$, and the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. Since the table shows values of x and their corresponding values of gx , substituting values of x and gx in the equation $gx = \frac{fx}{x+3}$ can be used to define function f . Using the first pair of values from the table, $x = -27$ and $gx = 3$, yields $3 = \frac{f-27}{-27+3}$, or $3 = \frac{f-27}{-24}$. Multiplying each side of this equation by -24 yields $-72 = f-27$, so the point $-27, -72$ is on the graph of $y = fx$. Using the second pair of values from the table, $x = -9$ and $gx = 0$, yields $0 = \frac{f-9}{-9+3}$, or $0 = \frac{f-9}{-6}$. Multiplying each side of this equation by -6 yields $0 = f-9$, so the point $-9, 0$ is on the graph of $y = fx$. Substituting $-27, -72$ and $-9, 0$ for x_1, y_1 and x_2, y_2 , respectively, in the formula $m = \frac{y_2 - y_1}{x_2 - x_1}$ yields $m = \frac{0 - (-72)}{-9 - (-27)}$, or $m = 4$. Substituting 4 for m in the equation $fx = mx + b$ yields $fx = 4x + b$. Since $0 = f-9$, substituting -9 for x and 0 for fx in the equation $fx = 4x + b$ yields $0 = 4(-9) + b$, or $0 = -36 + b$. Adding 36 to both sides of this equation yields $36 = b$. It follows that 36 is the y -coordinate of the y -intercept 0, b of the graph of $y = fx$. Therefore, the y -intercept of the graph of $y = fx$ is 0, 36.

Choice B is incorrect. 12 is the y -coordinate of the y -intercept of the graph of $y = gx$.

Choice C is incorrect. 4 is the slope of the graph of $y = fx$.

Choice D is incorrect. -9 is the x -coordinate of the x -intercept of the graph of $y = fx$.

Q6**C**

C. $qx = 140.55^x$

Choice C is correct. Since the value of qx decreases by a fixed percentage, 45%, for every increase in the value of x by 1, the function q is a decreasing exponential function. A decreasing exponential function can be written in the form $qx = a1 - \frac{p}{100}^x$, where a is the value of q_0 and the value of qx decreases by $p\%$ for every increase in the value of x by 1. If $q_0 = 14$, then $a = 14$. Since the value of qx decreases by 45% for every increase in the value of x by 1, $p = 45$. Substituting 14 for a and 45 for p in the equation $qx = a1 - \frac{p}{100}^x$ yields $qx = 141 - \frac{45}{100}^x$, which is equivalent to $qx = 141 - 0.45^x$, or $qx = 140.55^x$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. For this function, the value of qx increases, rather than decreases, by 45% for every increase in the value of x by 1.

B. 6

Choice B is correct. It's given that a quadratic function models the projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. It follows that an equation representing the model can be written in the form $fx = ax - h^2 + k$, where fx is the projectile's estimated height above the ground, in meters, x seconds after the launch, a is a constant, and k is the maximum height above the ground, in meters, the model estimates the projectile reached h seconds after the launch. It's given that the model estimates the projectile reached a maximum height of 51.1 meters above the ground 3 seconds after the launch.

Therefore, $k = 51.1$ and $h = 3$. Substituting 51.1 for k and 3 for h in the equation $fx = ax - h^2 + k$ yields $fx = ax - 3^2 + 51.1$. It's also given that the model estimates that the projectile was launched from an initial height of 7 meters above the ground. Therefore, when $x = 0$, $fx = 7$. Substituting 0 for x and 7 for fx in the equation $fx = ax - 3^2 + 51.1$ yields $7 = a(0) - 3^2 + 51.1$, or $7 = 9a + 51.1$. Subtracting 51.1 from both sides of this equation yields $-44.1 = 9a$. Dividing both sides of this equation by 9 yields $-4.9 = a$. Substituting -4.9 for a in the equation $fx = ax - 3^2 + 51.1$ yields $fx = -4.9x - 3^2 + 51.1$. Therefore, the equation $fx = -4.9x - 3^2 + 51.1$ models the projectile's height, in meters, above the ground x seconds after it was launched. The number of seconds after the launch that the model estimates that the projectile will return to a height of 7 meters is the value of x when $fx = 7$. Substituting 7 for fx in $fx = -4.9x - 3^2 + 51.1$ yields $7 = -4.9x - 3^2 + 51.1$. Subtracting 51.1 from both sides of this equation yields $-44.1 = -4.9x - 3^2$. Dividing both sides of this equation by -4.9 yields $9 = x - 3^2$. Taking the square root of both sides of this equation yields two equations: $3 = x - 3$ and $-3 = x - 3$. Adding 3 to both sides of the equation $3 = x - 3$ yields $6 = x$. Adding 3 to both sides of the equation $-3 = x - 3$ yields $0 = x$. Since 0 seconds after the launch represents the time at which the projectile was launched, 6 must be the number of seconds the model estimates that the projectile will return to a height of 7 meters.

Alternate approach: It's given that a quadratic function models the projectile's height, in meters, above the ground in terms of the time, in seconds, after it was launched. It's also given that the model estimates that the projectile was launched from an initial height of 7 meters above the ground and reached a maximum height of 51.1 meters above the ground 3 seconds after the launch. Since the model is quadratic, and quadratic functions are symmetric, the model estimates that for any given height less than the maximum height, the time the projectile takes to travel from the given height to the maximum height is the same as the time the projectile takes to travel from the maximum height back to the given height. Thus, since the model estimates the projectile took 3 seconds to travel from 7 meters above the ground to its maximum height of 51.1 meters above the ground, the model also estimates the projectile will

take 3 more seconds to travel from its maximum height of 51.1 meters above the ground back to 7 meters above the ground. Thus, the model estimates that the projectile will return to a height of 7 meters 3 seconds after it reaches its maximum height, which is 6 seconds after the launch.

Choice A is incorrect. The model estimates that 3 seconds after the launch, the projectile reached a height of 51.1 meters, not 7 meters.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8

3.44

Accept also: 86/25

The correct answer is 3.44. It's given that $f_0 = 86$ and that for each increase in x by 1, the value of f_x decreases by 80%. Because the output of the function decreases by a constant percentage for each 1-unit increase in the value of x , this relationship can be represented by an exponential function of the form $f_x = ab^x$, where a represents the initial value of the function and b represents the rate of decay,

expressed as a decimal. Because $f_0 = 86$, the value of a must be 86. Because the value of f_x decreases by 80% for each 1-unit increase in x , the value of b must be $(1 - 0.80)$, or 0.2.

Therefore, the function f can be defined by $f_x = 86 \cdot 0.2^x$. Substituting 2 for x in this function yields $f_2 = 86 \cdot 0.2^2$, which is equivalent to $f_2 = 86 \cdot 0.04$, or $f_2 = 3.44$. Either 3.44 or 86 / 25 may be entered as the correct answer.

Alternate approach: It's given that $f_0 = 86$ and that for each increase in x by 1, the value of f_x decreases by 80%. Therefore, when $x = 1$, the value of f_x is $(100 - 80)\%$, or 20%, of 86, which can be expressed as 17.2. Since $0.2 \cdot 86 = 17.2$, the value of f_1 is 17.2. Similarly, when $x = 2$, the value of f_x is 20% of 17.2, which can be expressed as 3.44. Since $0.2 \cdot 17.2 = 3.44$, the value of f_2 is 3.44. Either 3.44 or 86 / 25 may be entered as the correct answer.

Q9**B**

B. $fx = -3^x + 5$

Choice B is correct. The graph of $y = f(x) + 4$ is shown, and the graph passes through the point $(0, 8)$. Substituting 0 for x and 8 for y in this equation yields $8 = f(0) + 4$. Subtracting 4 from both sides of this equation yields $4 = f(0)$. Therefore, when $x = 0$, the value of $f(x)$ is 4 . Each of the given choices is in the form $fx = -3^x + k$, where k is a constant. Substituting 0 for x and 4 for $f(x)$ in this equation yields $4 = -3^0 + k$, or $4 = -1 + k$. Adding 1 to both sides of this equation yields $5 = k$. Substituting 5 for k in the equation $fx = -3^x + k$ yields $fx = -3^x + 5$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10**C**

C. $fx = 1282^{x-1}$

Choice C is correct. For the form of the function in choice C, $fx = 1281.6^{x-1}$, the value of $f(1)$ can be found as 1281.6^{1-1} , which is equivalent to 1281.6^0 , or 128 . Therefore, $k = 128$, which is shown in $fx = 1281.6^{x-1}$ as the coefficient.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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